

Supplier Data Request Package (RFI Templates)

Project: Fully underground civil shelter, private plot, Rusanivski Sady, Kyiv Purpose: Four ready-to-send Requests for Information (RFI), one per supplier category. Each asks the supplier for their own product data — it does not specify a final design, and none of the figures below (opening sizes, airflow, concrete class) are to be treated as confirmed design values. They are the client's current indicative concept, explicitly labeled as such, so suppliers can propose real products against something concrete. All four are subject to revision once the licensed structural engineer assigns a protection class (see Design Basis & Open Issues Register, item OI-04) and completes calculations (OI-05).

RFI-1: Protective / Sealed Entrance Door

To: [door supplier name] From: [client name / representative] Subject: Data request — protective and sealed doors for a private underground civil shelter

We are planning a fully underground civil shelter and are gathering supplier data before finalizing the design. Please treat all figures below as indicative only — we need your product's actual specifications so our engineer can finalize the opening and embedded parts around your real product, not the other way around.

Indicative opening (subject to change): approx. 900 mm wide rough opening; wall thickness approx. 300 mm reinforced concrete.

Please provide for your standard product range suitable for a private underground shelter:

- 1 Available protection classes / rated overpressure (kPa) and the standard(s) each is certified against.
 - 2 Door leaf and frame dimensions, and the exact rough-opening / embedded frame dimensions your product requires.
 - 3 Weight of the complete assembly (for lintel/jamb design).
 - 4 Anchoring / embedded-parts requirements (bolt pattern, embedment depth, required concrete cover).
 - 5 Whether you offer both an outer (protective/blast-rated) door and an inner airtight/sealed door as a matched pair, and their combined installation depth requirement (for our vestibule sizing).
 - 6 Sealing/gasket system and expected service life / maintenance interval.
 - 7 Lead time and current indicative pricing for budgeting purposes only (not a purchase order).
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RFI-2: Emergency Exit Hatch

To: [hatch supplier name] From: [client name / representative] Subject: Data request — emergency exit hatch for a private underground civil shelter

Indicative opening (subject to change): approx. 800 × 800 mm clear opening through a 400 mm reinforced-concrete slab, at the top of a vertical escape shaft.

Please provide:

- 1 Standard hatch sizes available and the exact rough-opening dimension each requires.
- 2 Frame/embedded-part details (steel frame dimensions, anchoring into concrete, required cover).

- 3 Rated load (pedestrian / soil surcharge above, if buried under landscaping) and any waterproofing/sealing certification.
 - 4 Locking/release mechanism suitable for emergency egress from inside, and whether it can be operated by a single person.
 - 5 Recommended ladder/rung system compatible with your hatch, if you offer one, or clearance dimensions needed to fit a third-party ladder.
 - 6 Lead time and current indicative pricing for budgeting purposes only.
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RFI-3: Ventilation Equipment

To: [ventilation supplier name] From: [client name / representative] Subject: Data request — ventilation for a private underground civil shelter, 4 occupants

We would like your recommendation, not ours — please size equipment based on your own methodology for a sealed underground shelter for 4 occupants, and tell us what you need from us to do that (room volume, occupancy schedule, whether NBC/CBRN filtration is required, etc.).

Please provide, for both a basic mechanical-ventilation option and a filtered/NBC-capable option:

- 1 Your recommended airflow rate and the standard/guideline you size it against.
 - 2 Equipment list (fan(s), filters, valves — including blast/overpressure valves if relevant to a filtered option), dimensions, and power consumption of each.
 - 3 Duct/sleeve diameters required through the concrete envelope, for how many penetrations (supply/exhaust), and any manual backup (hand-crank) options.
 - 4 Required sensors/interlocks you recommend (CO₂, CO, smoke, or others) and whether these are included or third-party.
 - 5 Maintenance interval and filter replacement schedule/cost.
 - 6 Lead time and current indicative pricing for budgeting purposes only.
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RFI-4: Concrete Supply

To: [concrete plant / supplier name] From: [client name / representative] Subject: Data request — waterproof concrete mix for a private underground civil shelter

We are unable to apply external membrane waterproofing on this structure, so we are exploring an integral waterproof-concrete ("white box") approach and need your plant's actual mix design and guidance, not a generic assumption.

Please provide:

- 1 Your standard mix designs available at strength/grade in the general range of C35/45 (or your closest equivalent), with water-penetration (W-class or equivalent) and freeze-thaw (F-class or equivalent) ratings you can certify.
- 2 Any integral waterproofing admixture (e.g., crystalline) systems you offer or recommend for a fully buried structure without membrane waterproofing, and technical literature/certification for them.
- 3 Waterstop and joint-sealing products you supply or recommend for construction/movement joints, compatible with your mix.
- 4 Pumping/delivery capability to a private plot in Rusanivski Sady, Kyiv (truck-mixer access, pump availability), and typical batch sizes.

- 5 Lead time from order to delivery, and current indicative pricing per m³ for budgeting purposes only (final quantities to be confirmed by the structural engineer's issued-for-construction drawings — not by this request).
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Notes for all four RFIs

- Do not commit to a purchase order based on these responses alone.
- Forward all supplier responses to the licensed structural engineer (see Engineer Handover Sheet, Doc.02) — embedded parts, openings, and reinforcement around openings must be finalized against the actual product dimensions received here, not the indicative figures above.
- Track responses in the Design Basis & Open Issues Register (Doc.01), closing OI-06 (door/hatch/ventilation) and OI-07 (concrete) once supplier data is in hand and reviewed by the engineer.